Assignment10 – Natural Language

Given: June 24 Due: June 29

Problem 10.1 (Ambiguity)

1. Explain the concept of ambiguity of natural languages.

Solution: Ambiguity is the phenomenon that in natural languages a single utterance can have multiple meanings.

---

2. Give two examples of different kinds of ambiguity and explain the readings.

Solution: Here are some examples

- "bank" can be a financial institution or a geographical feature.
- In "I saw her duck" the word "duck" can be a verb or a noun.
- "Time flies like an arrow" could be about the preferences of special insects ("time flies") or about the fact that time passes quickly – e.g. in an exam.
- In "Peter saw the man with binoculars", it could be Peter who is using binoculars, or it could be that Peter saw "the man" who had "binoculars".

---

an algorithm A that takes a piece of text like:

"El presidente habló con la prensa."

Problem 10.2 (Language Identification)

You are given an English, a German, a Spanish, and a French text corpus of considerable size, and you want to build a language identification algorithm A for the EU administration. Concretely A takes a string as input and classifies it into one of the four languages $\ell^* \in \{English, German, Spanish, French\}$. The prior probability distribution for the strings being English/German/Spanish/French, is $\langle 0.4, 0.2, 0.15, 0.15\rangle$.

How would you proceed to build algorithm A? Specify the general steps and give/derive the formula for computing ℓ given a string $\mathbf{c}_{1:N}$.

Take a large corpus (lots of text).

Count how often each trigram appears.

Convert counts into probabilities.

Solution:

1. Build a trigram trigram language model $\mathbf{P}(c_i|\mathbf{c}_{i-2:i-1}, \ell)$ for each candidate language ℓ by counting trigrams in an ℓ-corpus.

"How likely is the sequence 'th' in English?"

"How likely is 'sch' in German?"

"How likely is 'que' in Spanish?"

2. Apply Bayes' rule and the Markov property to get the most likely language:

$$\begin{aligned}
\ell^* &= \operatorname*{argmax}_{\ell} (P(\ell \mid \mathbf{c}_{1:N})) && \text{Choose the language that is most probable given the text.} \\
&= \operatorname*{argmax}_{\ell} (P(\ell) \cdot P(\mathbf{c}_{1:N} \mid \ell)) && \text{prior prob * prob that text was generated by trigram model of language l (using bayes rule)} \\
&= \operatorname*{argmax}_{\ell} (P(\ell) \cdot (\prod_{i=1}^{N} P(c_i \mid \mathbf{c}_{i-2:i-1}, \ell))) && \text{using markov assumption}
\end{aligned}$$

A trigram is a sequence of three characters (or sometimes three words).

Problem 10.3 (Language Models)

Example (characters):

"th e"

"he r"

"er e"

Example (words):

"I love cats"

"love cats a"

"cats a lot"

Character n-gram models
Unigrams (1 character)

Bigrams (2 characters)

Trigrams (3 characters)

4-grams, 5-grams, etc.

1. How can we obtain a trigram model for a natural language? Explain the probability distribution involved.

Solution: We need a corpus of words over L. Then we count how often each trigram occurs in it and use that to estimate the probability distribution $P(T = t)$ of trigrams t.

---

2. Explain informally how we can use trigram models to identify the language of a document D.

Solution: We build a trigram model for each candidate language. Then we use each model to compute the probability of D occurring in that language. We choose the language with the highest probability.

---

3. Explain briefly what named entity recognition is.

Solution: The task of finding, in a text, names of things and deciding what class they belong to.

"Barack Obama visited Berlin in July."
NER identifies:

"Barack Obama" PERSON
"Berlin" LOCATION
"July" DATE

The probability of the next character depends only on the previous two.

This is the Markov assumption.

$P(c_i \mid c_{i-2}, c_{i-1})$